SRIDURGESH RAVICHANDRAN

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SUMMARY

Master of Computing (AI) candidate at NUS with hands-on experience building and evaluating machine learning systems across foundation models, multimodal learning, graph learning, reinforcement learning, and AI for Science. Experienced in model training and fine-tuning, distributed GPU experimentation, end-to-end ML pipelines, and rigorous evaluation using PyTorch and modern deep learning frameworks. Strong background in both research and applied AI, with peer-reviewed publications and experience translating complex modelling ideas into practical, measurable systems

EDUCATION

National University of Singapore, Master of computing in AI **2025-2026**

SRM Institute of Science and Technology, **GPA**: 9.09 / 10.00
BTech with specialization in Artificial Intelligence and Machine Learning.

Relevant Coursework at NUS

| Graph Machine Learning (graph models and high level recommendation system design) | Trust worthy machine learning (Model attacks and defense) | Neural network and deep learning (Hands on work with LLM) | Knowledge discovery and data mining (data science and analytics) | AI planning and decision making (Reinforcement learning) | Neural Network and Deep Learning 2 (exploring High performance computing with optimization kernels on GPU like flash attention , kernel fusion , KV cache and efficient training and inference) |

Recent Work Experience

AI Singapore
Description: Part time student assistant assisting in data curation for training multi lingual LLM SEALion

The Yang Zhang Lab (Dissertation)
Description : Lab member at Professor Yang Zhang lab working in Biological Protein RNA modelling .
Lab page : https://zhanggroup.org/people/

PROJECTS

1) **iPotNET - Physics Informed AI for Microscopy Inverse problem**
   Github: AI for Microscopy
   Description:Developed a physics-informed deep learning framework (SwinUNETR) to reconstruct atomic potentials from noisy STEM images. Engineered an 11-channel metadata fusion pipeline that solves the inverse scattering problem with >54 dB PSNR, outperforming traditional iDPC methods by resolving atomic columns in thick (150nm), tilted samples.Generated a "Gold Standard" training dataset using DFT (CP2K) and Multislice simulations (abTEM). **In Review (MLST Journal) .**

2) **Large Language Model Training, Fine-Tuning & Evaluation**
   Description: Fine-tuned **Qwen2.5-14B** on a medical corpus using **LoRA**, with synthetic QA data generated by Mistral-7B-Instruct and filtered using accuracy/style quality thresholds to reduce hallucinations. Trained **Qwen3** from scratch on **Alien-C4** using the **Muon optimizer** and benchmarked it against Adam in a distributed **A100/SLURM** setup, analyzing convergence, loss stability, and throughput. Additionally fine-tuned **Llama-2-7B** on 15K Databricks instruction samples using **LoRA/PEFT** with multiple rank configurations and **DDP**, evaluating performance using **AlpacaEval and MT-Bench** with absolute and pairwise assessments.

3) **Graph Integrated LLM**
   Description: Developed a hybrid Graph–LLM "Glass-Box Toxicity Predictor" combining GNNs (GINE) with DeepSeek-R1 to generate interpretable molecular toxicity predictions. Integrated adaptive RAG, graph token projection, multi-head loss (LM loss + graph-head + LLM label loss), and scaffold-split benchmarking,with fully interpretable reasoning traces.

4) **Training LLM with DPO and IPO with exploring Quantization methods**
   Description: Trained language model with DPO and IPO policy optimization replacing RLHF using Anthropic HH-RLHF dataset and explored different quantization techniques like Symmetric , Asymmetric , perchannel , recursive model quantization and smooth quantization with SNR analysis and calibration clipping .

5) **HDB Resale Price Prediction using Geospatial & Ensemble Learning**
   Description: **Python, XGBoost, LightGBM, CatBoost, SHAP, Optuna** | GitHub Engineered temporal, categorical, and **geospatial accessibility features** from 162K+ Singapore housing transactions and auxiliary MRT, school, mall, and hawker-centre datasets.Built and optimized a **6-model stacking ensemble**, attaining **SGD 24.9K test RMSE** and **98.2% validation R^2**, with SHAP-based interpretation of key housing-price drivers.

6) **RL in Game environment**

Description: **RL pipeline** in PyTorch using a custom Proximal Policy Optimization (PPO) Actor Critic architecture to solve partially observable, sparse-reward grid environments. **Designed dual neural network backbones**, leveraging CNNs for dynamic spatial navigation and LSTMs to overcome amnesia in memory-bottlenecked tasks. **Optimized for sample efficiency and training stability**, implementing batched Generalized Advantage Estimation (GAE) and explicit KL-divergence penalties to prevent catastrophic forgetting during high-throughput parallel rollouts.

7) **Privacy Attacks & Defenses in Machine Learning Systems - AI Safety**

Description: Studied privacy leakage in ML systems through **membership inference, intersection attacks, k-anonymity failures, database reconstruction, and model inversion**. Evaluated inversion attacks on **ArcFace embeddings** under unprotected and privacy-protected settings using attack success rate and FID, and analyzed **differential privacy** as a randomized defense against individual-level information leakage even when adversaries possess auxiliary information. **Technical Article:** [Article Link

8) **Adversarial Robustness, Backdoors & Federated Learning Security - AI Safety**

Description: **Description**: Evaluated security vulnerabilities in deep learning systems through **targeted Carlini–Wagner attacks, backdoor trigger reverse engineering, and federated-learning poisoning attacks**. Implemented embedding-space adversarial attacks against **FaceNet**, recovered hidden backdoor triggers using gradient-based optimization on MNIST and CelebA settings, and applied a **Rand-Eigen-style defense** to reduce malicious-update impact in federated training while comparing clean, poisoned, and defended learning dynamics.

PUBLICATIONS

1. SafeV-JEPA: World-Model Plausibility Lets You Measure Synthetic-Video Risks . Accepted in **ACMMM 2026** Brazil Co-First-authored under Prof.Mohan Kankanhalli .
2. Mechanistic Evidence Routing in Transformers: Margin-Gated Intervention for Retrieval-Conflict Hallucination **AACL-IJCNLP 2026 Main Track**, Solo Author
3. Breast Cancer Diagnosis with Histopathology Images Using Adaptive Feature Extraction and Machine Learning Model. Published in **ICDIC 2024** First-authored. Link: https://ieeexplore.ieee.org/document/10810857
4. Early Detection of Diabetic Foot Ulcer with Multi-Modal Images Using Federated Learning. Published in **ICMSCI 2024** Co First-authored. Link: https://ieeexplore.ieee.org/document/10894410.
5. Feature Controlled Synthetic Data Generation For Medical Imaging Using CGAN . Nepal **ICICT 2025** Co First-authored.
6. Link: https://ieeexplore.ieee.org/document/11005351

CURRENT THESIS WORKS

1. **Protein mutation interaction and optimization for enhancing fitness landscape the lens of epistasis (in progress)**
2. **RNA 3d structure prediction modelling with retrieval based RNA Language model (currently in progress)**

TECHNICAL SKILLS

Programming: Python, C++, SQL, Bash

Systems & Infrastructure: CUDA, PyTorch DDP, SLURM, multi-GPU/distributed training, GPU cluster workflows, mixed-precision training,Linux, Git/GitHub, Docker, Conda

Model Stacks: PyTorch, Hugging Face, scikit-learn , Triton basics .

Foundation & Generative Models: LLM training/fine-tuning PEFT/LoRA, DPO/IPO, RAG, quantization,

synthetic data generation MML , Physics neural operator and PINN based modelling..

ML & Data: Decision tree , clustering , ensembling modelling like XGBoost, LightGBM, CatBoost, NumPy, pandas, Optuna, SHAP

Evaluation & Research: Benchmarking, ablation studies, robustness and OOD evaluation, model comparison, error analysis

Scientific ML: CP2K, abTEM

Applications: FastAPI, LangChain , Custom Agent Harness design .